MIL-C-10872E
30 September 1985
SUPERSEDING
MIL-C-10872D
24 May 1974

MILITARY SPECIFICATION

CAP, FLUE, 4-INCH, CYLINDRICAL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers one type and size of steel flue cap of cylindrical design, with guy lines.

2. APPLICABLE DOCUMENTS

* 2.1 Government documents. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this document to the extent specified herein.

SPECIFICATIONS

FEDERAL

T-C-571	-	Cord, Cotton; General and Special Purpose, Sash and Venetian Blind
QQ-S-698	-	Steel, Sheet and Strip, Low-Carbon
QQ-W-461	-	Wire, Steel, Carbon, (Round Bare and Coated)
PPP-B-636	-	Boxes, Shipping, Fiberboard
PPP-T-45	-	Tape, Gummed, Paper, Reinforced and Plain, for Sealing and Securing
PPP-T-60	-	Tape; Packaging, Waterproof

Beneficial comments (recommendations, additions, deletions) and any pertinent data that may be of use in improving this document should be addressed to U.S. Army Natick Research and Development Center, Natick, MA 01760-5014, by using the self-addressed Standardization Document Improvement Proposal (DD form 1426) appearing at the end of this document or by letter.

FSC 4520

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MILITARY

MIL-P-116 - Preservation, Methods Of
MIL-B-121 - Barrier Material, Greaseproofed, Waterproofed, Flexible

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads

DRAWING

U.S. ARMY NATICK RESEARCH AND DEVELOPMENT CENTER

2-9-89 - Cap, Flue, 4-Inch, Cylindrical

(Copies of documents required by manufacturers in connection with specific acquisition functions should be obtained from the procuring activity or as directed by the contracting officer.)

* 2.2 Other publications. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this document to the extent specified herein.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951- Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

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2.3 Order of precedence. In the event of conflict between the text of this document and the references cited herein, the text of this document shall take precedence.

3. REQUIREMENTS

3.1 First article. When specified, a sample shall be subjected to first article inspection (see 4.3, 6.2, and 6.3).

* 3.2 Materials. Recycled materials may be used (see 6.4).

* 3.2.1 Steel, sheet. Sheet steel shall be hot or cold rolled, drawing quality, 1009 through 1020, as rolled or No. 1 matte finish in accordance with QQ-S-698.

3.2.2 Steel wire. Steel wire shall conform to finish 1, annealed temper, any composition of QQ-W-461.

3.2.3 Rivets. The rivets used shall be 1 1/2-pound steel tinners rivets in accordance with manufacturer's commercial practice.

3.2.4 Cord, cotton, solid, braided, 1/8-inch diameter. The cord used to fabricate the guy lines shall conform to type I, class 3, size 4 of T-C-571.

3.3 Construction. The cap shall conform to the requirements of Drawing 2-9-89 and as specified herein.

3.3.1 Welding surface pretreatment. Prior to spot welding as shown on Drawing 2-9-89, all surfaces to be spot welded shall be thoroughly cleaned.

3.3.2 Riveting. The rivets shall be inserted with the head on the inside of the bottom band. The rivet head shall be flush with the inside surface of the lower inner band, by indentation of the band. The rivets shall be securely set and neatly headed on the outside of the upturned end of the spacers as shown on Drawing 2-9-89.

3.3.3 Loop. After assembly of the loop straps on the cap, the loops shall swing freely of their own weight.

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3.3.4 Sleeves, oval. Oval sleeves shall be tightly squeezed to form loops at the ends of the extension cable for the cable retainer loop and guy cord as shown on Drawing 2-9-89.

3.3.5 Guy lines. The three guy lines shall be individually tied to the three guy line retainer loops with double overhand knots as illustrated in Drawing 2-9-89.

3.4 Finish. Sheet steel shall be oxidized in the sheet to produce a uniform, tightly adherent, dark blue-black color.

3.5 Marking for identification. Marking for identification shall be done by indentation of the cap with the letters "US," the manufacturer's name, number of contract, and year of manufacture in the size and approximate location shown on Drawing 2-9-89. When the cap, except for guy lines, is fabricated by a manufacturer other than the prime contractor, the marking specified above shall be supplemented by the words "made by" and followed by the actual fabricator's name.

3.6 Workmanship. Each flue cap shall be rigid after assembly, clean, and free of rust, burrs, sharp edges, grooves, laps, or projections. All welds shall be smooth, sound, and free from pits, flashes, burn through, and undercut.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the document where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.3).
2. Quality conformance inspection (see 4.4).

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4.3 First article inspection. When a first article is required (see 6.2), the flue cap shall be examined as specified in 4.4.2.1 and 4.4.2.2. Any nonconformance shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase documents.

4.4.2 End item inspection. The inspection lot shall be all flue caps offered for inspection at one time. The sample unit shall be one completed flue cap.

4.4.2.1 Visual inspection. The flue cap shall be examined for the defects in table 1. The inspection level shall be II with an acceptable quality level (AQL) of 2.5 for major defects and 6.5 for total defects, expressed in terms of defects per hundred units.

TABLE I. Classification of defects

Examine	Defect	Classification	
		Major	Minor
Finish	Not oxidized to produce a uniform tightly adherent, dark blue-black color, or any unoxidized areas	101	
Construction and workmanship	Any sharp edge, projection, or burr	102	
	Any component malformed, dented, cracked, fractured, punctured, or is wrong type, material, or design	103	
	Any component damaged, buckled, warped, or bent out of shape	104	
	Bottom edge of bottom band not flared	105	
	Less than three spot welds on any band joint	106	

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TABLE I. Classification of defects - Continued

Examine	Defect	Classification	
		Major	Minor
Construction and workmanship (continued)	Outside band not concentric with inside band, or bands not formed at joint to make a flush inside surface (see section B-B of Drawing 2-9-89)	107	
	Grooves or laps		201
	Evidence of corrosion	108	
	Surfaces not clean (foreign material)		202
	Bottom and outside bands not beaded	109	
	Any weld omitted, not fused, cracked, or fractured	110	
	Any weld undercut or burned through		203
	Welds not free from pits and flash		204
	Any joint not held securely by spot welding	111	
	Spot welding not in locations shown on drawing	112	
	Any component missing	113	
	Retainer loop binds (does not swing freely of its own weight)		205
	Any rivet not peened, rivet head broken off, or rivet is loose	114	
	Any rivet peened on inside of bottom band, not neatly peened	115	
	Any rivet not flush on inside of bottom band	116	
	Guy line cord cut or abraded	117	
	Guy line cord not solid braided, or not tied with double overhand knot (as specified)	118	
	Guy line cord is glazed, excessively greasy, tacky, oily, or stiff		206

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TABLE I. Classification of defects - Continued

Examine	Defect	Classification	
		Major	Minor
Identification marking	Omitted, or not indented or die stamped, illegible, incomplete, incorrect, in wrong location, or wrong size		207

4.4.2.2 Dimensional examination. Inspection shall be made of the cap for compliance with dimensional requirements. Any deviation from the specific requirements shall be classified as a defect. The inspection level shall be S-2 with an AQL of 4.0 defects, expressed in terms of percent defective

4.4.3 Packaging inspection. An inspection shall be made to determine that preservation, packing, and marking comply with the section 5 requirements. Defects shall be scored as specified in table II. The lot size shall be the number of shipping containers in the inspection lot. The sample unit shall be one shipping container fully packed. The inspection level shall be S-2 with an AQL of 4.0 defects, expressed in terms of defects per hundred units.

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TABLE II. Packaging inspection

Examine	Defect
Markings	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application
Preservation	Preservative improperly applied or missing
Materials	Component missing, damaged, or otherwise defective
Content	Number per container is more or less than required

* 4.4.4 Palletization inspection. An inspection shall be made to determine that the palletization complies with the section 5 requirements. Defects shall be scored in accordance with table III. The lot size shall be the number of palletized unit loads in the inspection lot. The sample unit shall be one palletized unit load fully packaged. The inspection level shall be S-1 and the AQL shall be 6.5, expressed in terms of defects per hundred units.

TABLE III. Palletization inspection

Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirements
Palletization	Pallet pattern not as specified Interlocking of loads not as specified Load not bonded with required straps as specified
Weight	Exceeds maximum load limits
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application

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5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial as specified (see 6.2).

5.1.1 Level A.

* 5.1.1.1 Preservative application . Each cap shall be cleaned, process C-1, thoroughly dried, and coated with type P-7 preservative in accordance with MIL-P-116. Each attached guy line shall be folded to a small, neat bundle and tied in the center with cotton, sisal, or jute cord or twine or fastened with tape. The three bundled guy lines shall then be brought together and wrapped in barrier material conforming to type I or II, grade A of MIL-B-121, and the wrap secured with cotton, sisal, or jute cord or twine; or one-inch minimum width tape conforming to PPP-T-60, or type III, grade A of PPP-T-45. Not more than six inches of guy lines shall be exposed outside the wrapping. Each guy line retainer lock shall be folded against the bottom board and the wrapped guy lines placed within the cap. Each cap shall then be wrapped in greaseproof barrier material and closure effected with cord, twine, or tape as specified herein for the guy lines.

5.1.1.2 Unit packing. Each cap, prepared as specified in 5.1.1.1, shall be unit packed in a snug-fitting fiberboard box conforming to style RSC (end-opening), type CF, variety SW, or type SF, class domestic of PPP-B-636. Closure shall be in accordance with method II of the appendix of PPP-B-636.

5.1.2 Commercial. The caps shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B, or Commercial as specified (see 6.2).

5.2.1 Level A packing. Thirty caps, preserved as specified in 5.1, shall be packed upright, in the most compact manner, in a snug-fitting fiberboard shipping container conforming to style RSC, V2s of PPP-B-636. Each shipping container shall be closed in accordance with method III, waterproofed in accordance with method V, and reinforced as specified in the appendix of PPP-B-636.

5.2.2 Level B packing. Thirty caps, preserved as specified in 5.1, shall be packed upright, in the most compact manner, in a snug-fitting fiberboard shipping container conforming to style RSC, type CF, variety SW, or type SF, class domestic, grade 275 of PPP-B-636. Closure shall be in accordance with method II of the appendix of PPP-B-636.

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5.2.2.1 Weather-resistant fiberboard containers. When specified (see 6.2), the shipping container shall be a V3c, V3s or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in the appendix of PPP-B-636.

* 5.2.3 Commercial packing. Caps, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

* 5.3 Palletization. When specified (see 6.2), caps packed as specified in 5.2, shall be palletized on a four-way entry pallet in accordance with load type Ia of MIL-STD-147. Pallet type shall be type I (four-way entry), type IV or type V in accordance with MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L or film bonding means O or P. Pallet patterns shall be in accordance with the appendix of MIL-STD-147. Interlocking of loads shall be effected by reversing the pattern of each course. If the container is of a size which does not conform to any of the patterns specified in MIL-STD-147, the pallet pattern used shall first be approved by the contracting officer.

* 5.4 Marking. In addition to any special marking required by the contract or purchase order, unit packs, shipping containers, and palletized unit loads shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

6. NOTES

6.1 Intended use. The flue cap is used with liquid fuel burning devices having stove pipe or flue terminations ending in 4-inch stove pipe sizes.

* 6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this document.
b. When a first article is required (see 3.1).
c. Selection of applicable levels of preservation and packing (see 5.1 and 5.2).
d. When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).
e. When palletization is required (see 5.3).

* 6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all acquisition instruments regarding arrangements for inspection and approval of the first article.

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* 6.4 Recycled material. The contractor is encouraged to use recycled material when practical, provided that it meets the requirements of this document (see 3.2).

6.5 Changes from previous issue. The margins of this document have been marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written irrespective of the marginal notations and relationship to the last previous issue.

Custodian:

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Preparing activity:

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User activity:

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-10872E	Cap, Flue, 4-Inch, Cylindrical

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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NATICK OP, 1 Apr 84

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